PulseIQ - Phase 1 Execution Report

This report summarizes the implementation and validation of PQ-01 through PQ-04. Results reflect local execution evidence. Outstanding items are identified separately.

1. Scope and implementation

Task	Implementation and verified result
PQ-01 - ETL and contracts	Processed 29 CSV inputs and 62,770 rows. Accepted 62,732 rows and quarantined 38; row accounting passed. Schema drift is fail-closed. The 58 clean/rejected CSV outputs were byte-identical across two consecutive runs.
PQ-02 - Warehouse	Built 25 warehouse tables. All 25 table reconciliations matched. The project records 39 foreign-key constraints and zero orphan rows in five representative joins.
PQ-03 - Data quality	Executed Python and SQL checks. All 14 SQL checks were exercised through mutation tests on a separate test database; evidence is in outputs/dq_mutation_test_evidence.csv.
PQ-04 - Metrics	All 28 metric views executed successfully. Six independent hand-checks matched their corresponding metric views. M28 is marked not yet measurable because overlapping actuals are unavailable.

2. Data-quality results

Baseline checks were run against raw source data. FAIL indicates a detected data-quality finding; it does not by itself mean the detector failed or that the clean warehouse contains the defect.

Check	Result	Findings	Check	Result	Findings
DQ-C01 Payroll/ledger	FAIL	5	DQ-C08 Expense arithmetic	FAIL	1
DQ-C02 Ledger balanced	PASS	0	DQ-C09 Task dates	FAIL	20
DQ-C03 Unique payroll	FAIL	4	DQ-C10 Leave dates	FAIL	2
DQ-C04 Orphan employees	FAIL	2	DQ-C11 AI feed freshness	FAIL	410
DQ-C05 Attendance after exit	FAIL	4	DQ-C12 History overlap	FAIL	1
DQ-C06 Unpaid leave	FAIL	1	DQ-C13 Pipeline freshness	PASS	0
DQ-C07 Invoice arithmetic	FAIL	1	DQ-C14 Minimum group size	PASS	0

All 11 planted issues are covered in docs/planted_issue_reconciliation.md. DQ-C11 includes five stale attrition predictions and 405 anomaly scores older than 24 hours relative to the run, associated with the static training snapshot. DQ-C14 passed with zero baseline violations; removing suppression in the mutation-test copy exposed 81 below-threshold groups.

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3. Metric verification and reconciliation

Six independent hand-checks matched the corresponding metric views. M19 uses ledger accounts 5001-5005 for payroll cost, while retaining payroll-file cost for reconciliation visibility.

Open issue - June 2026 payroll versus ledger: Payroll employer cost is INR 54,664,233.35, while ledger salary-related expense is INR 54,682,683.35. The unresolved difference is INR 18,450.00. The journal entry balances, but salary expense does not reconcile to payroll. No source row or ledger posting has been silently adjusted.

4. Automated validation and evidence

- 13 automated tests passed using python -m pytest.

- All 29 input tables and 62,770 rows were processed; 62,732 clean and 38 rejected rows reconcile exactly.

- All 25 warehouse table reconciliations matched.

- All 28 metric views and all 14 DQ SQL check IDs were verified; mutation-test results are recorded separately from the baseline.

- The original outputs/pulseiq.duckdb is the baseline. outputs/pulseiq_mutation_test.duckdb is a test-only database containing deliberate mutations.

5. Outstanding sign-off items

- Confirm the approved anomaly reason_code values with the project owner; an authoritative list has not yet been confirmed.

- Obtain an explanation for the June 2026 INR 18,450 payroll-to-ledger difference or secure formal acceptance of the exception.

- Complete final review and sign-off after these items are addressed or formally accepted.

Supporting evidence: README.md; docs/phase1_execution_report.md; docs/dq_gate_policy.md; docs/dq_test_matrix.md; docs/metric_hand_checks.md; docs/planted_issue_reconciliation.md; outputs/warehouse_row_reconciliation.csv; outputs/metric_hand_checks.csv; outputs/dq_mutation_test_evidence.csv.

Open Items and Clarifications Required

This section provides the details needed for owner confirmation. Both items remain pending; neither is represented as resolved.

1. June 2026 payroll-to-ledger reconciliation

A difference of INR 18,450.00 was identified between the payroll-based expected amount and the amount posted to ledger account 5001 for department D01.

Field	Value
Ledger line	JL-005614
Journal	JE-002094
Account / department	5001 / D01
Ledger amount	INR 1,98,91,524.18
Payroll-based expected amount	INR 1,98,73,074.18
Difference	**INR 18,450.00**

Status: Pending Finance review and confirmation of the appropriate accounting treatment. No unsupported adjustment has been made to the source data.

2. Anomaly reason_code allow-list

Eleven distinct values were observed in the supplied anomaly dataset: NO_ATTENDANCE, PATTERN_SHIFT, LONG_UNPAID_ABSENCE, STATISTICAL_OUTLIER, GROSS_ABOVE_NORM, NEGATIVE_NET_PAY, OVERTIME_ABOVE_CAP, ZERO_PAY, GROSS_BELOW_EXPECTED, FREQUENT_LATE, and PF_NOT_DEDUCTED.

Status: These are observed dataset values, not a verified official allow-list. The authoritative list was not available in the project files reviewed. Confirmation is required from the project owner or designated data-governance owner before final allow-list validation.

3. Required follow-up

- Finance to explain the INR 18,450.00 difference or provide an approved adjustment reference or formal exception acceptance.

- Project owner or data-governance owner to provide or approve the authoritative anomaly reason_code list.

- After confirmation, rerun the affected reconciliation and allow-list checks, update the evidence, and complete final sign-off.

The original source data and validation evidence have been preserved. No financial posting or official code list has been changed or represented as approved without owner confirmation.